<table>
<tr><td></td><td colspan="2">Marwadi University
Faculty of Engineering & Technology
Department of Information and Communication Technology</td></tr>
<tr><td>Subject: DIGITAL VLSI TESTING (01CT0629)</td><td colspan="2">Aim: Detection of both SA0 and SA1 faults on a VLSI circuit that represents 3–input EVEN function.</td></tr>
<tr><td>Experiment No: 2</td><td>Date:04-03-2026</td><td>Enrollment No: 92301733056</td></tr>
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Aim:

1. To understand stuck-at-0 and stuck-at-1 fault using 3-input EVEN function.
2. Implementation and visualization of stuck-at-0 and stuck-at-1 fault with the help of XOR gate simulator.
3. Detection of both SA0 and SA1 faults on a 3–input EVEN function logic circuit.

Simulation Platform: Virtual Lab

https://vlsi-nitk.vlabs.ac.in/exp/3-input-even-function/

Theory:

A fault model is an engineering model of something that could go wrong in the construction or operation of a piece of equipment, A stuck-at fault is a particular fault model used by fault simulators, Individual signals and pins are assumed to be stuck at Logical '1', '0' . For example, an input is tied to a logical 1 state during test generation to assure that a manufacturing defect with that type of behaviour can be found with a specific test pattern. The even function is equal to 1 when the number of 1’s in the input combination is even.
The complement of an odd function is an even function. The complement of an odd function (an even function) is obtained by replacing the output gate with an Exclusive-NOR gate.
Combinational circuit is a circuit in which we combine the different gates in the circuit, for example encoder, decoder, multiplexer and demultiplexer.

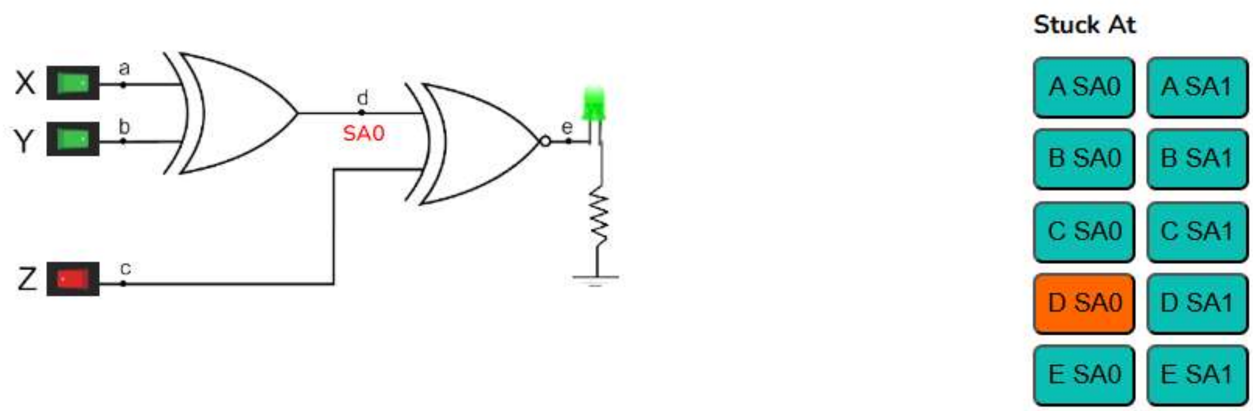
The characteristics of combinational circuits

1. The output of combinational circuit at any instant of time, depends only on the levels present at input terminals.
2. The combinational circuit do not use any memory. The previous state of input does not have any effect on the present state of the circuit.

3. A combinational circuit can have an n number of inputs and m number of outputs.

Simulation:
Paste your simulation result screen shot here:

Simulation 1



Simulation 2

X	Y	Z	Fault	Expected Output	Faulty Output
0	0	0	N/A	0	1
0	0	0	A SA0	1	1
0	0	1	A SA0	0	0
0	1	0	A SA0	0	0
0	1	1	A SA0	1	1
1	0	0	A SA0	0	1
1	0	1	A SA0	1	0
1	1	0	A SA0	1	0
1	1	1	A SA0	0	1
0	0	1	C SA0	0	1

Truth Table:

Input			Fault	Expected Output	Faulty Output
X	Y	Z			
0	0	0	N/A	0	1
0	0	0	A SA0	1	1
0	0	1	A SA0	0	0

Conclusion:

Post Lab Exercise

1. Electronic circuits designed to process one or more input signals and generate a defined output is ________________________ .

(A) Parallel circuits (B) Logic signals ←

(C) Logic gates (D) Series circuits

2. What is the purpose of a single stuck-at fault model in circuit testing?

(A) Pre-Manufacturing testing

(B) Post-Manufacturing Testing←

(C) Design Testing

(D) None of the above

3. Which of the following are universal gates?

(A) NAND, NOR ← (B) NAND, OR

(C) NOT, XOR (D) NONE OF THE ABOVE

4. When does the XOR gate produce a high output?

(A) When both inputs are 1

(B) When both inputs are 0

(C) When one input is 1 and the other is 0

(D) When both inputs are either 0 or 1

5. Output of E- XOR (Exclusive OR) gate is high when ______________________________ .

(A) When one input is low and the other is high

(B) Only when both inputs are low

(C) When both inputs are equal (either 0 or 1)

(D) Only when both inputs are high

6. The expression of an XOR gate is?

(A) A’.B+A.B’

(B) A.B+A’.B’

(C) A.B’

(D) A.A’+B.B’

7. Which of the following is the Boolean expression for an EX-NOR gate?

(A) A’.B

(B) A.B+A’.B’

(C) A’.B+A.B’

(D) A.A’+B.B

8. For a circuit with k lines, __________ single stuck-at faults are possible?

(A) k

(B) 2k

(C) k/2

(D) k^2

9. When logic gates are interconnected to form a logic network, it is referred to as a ______ logic circuit.

(A) Combinational

(B) Sequential

(C) Hardwires

(D) Systematic

10. What is the total number of distinct multiple stuck-at faults possible in a circuit with n lines?

(A) 3^n   (B) 3^n-1

(C) 2^n-1   (D) 2